

Riemann Zeros and the Trace Formula on $\mathrm{SO}_0(5, 2)(\mathbb{Z}) \backslash D_{IV}^5$

Class Number 1, Universal Representation, and a Precise Conjecture

Casey Koons (Independent Research)

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Abstract

We establish three arithmetic facts about the quadratic form $Q = x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - x_6^2 - x_7^2$ of signature $(5, 2)$: its class number is 1 (Milnor 1958), it represents all integers (Lagrange), and $\mathrm{Spin}(5, 2)$ satisfies strong approximation (Kneser 1966). The associated bounded symmetric domain $D_{IV}^5 = \mathrm{SO}_0(5, 2)/[\mathrm{SO}(5) \times \mathrm{SO}(2)]$ therefore has the simplest possible arithmetic structure: a unique lattice, one connected component, no genus complications.

We compute the Eisenstein series intertwining operators for the minimal parabolic of $\mathrm{SO}_0(5, 2)$ explicitly via the B_2 root system and show that the completed Riemann zeta function $\xi(s)$ appears in the c -function as ratios $\xi(s_\alpha)/\xi(s_\alpha + 1)$ over the four positive roots. We prove that on the standard unitary axis, these ratios are evaluated at real parts 1, 3, 4, 5 — not at $1/2$. The self-adjointness of the Bergman Laplacian therefore does not directly constrain ζ -zeros through the continuous spectrum.

We conjecture that the Arthur–Selberg trace formula for $\mathrm{SO}_0(5, 2)(\mathbb{Z}) \backslash D_{IV}^5$, with class number 1 on the geometric side and self-adjointness on the spectral side, forces all nontrivial zeros of $\zeta(s)$ onto $\mathrm{Re}(s) = 1/2$. The conjecture is stated precisely and the obstacles are identified honestly. We invite its resolution by specialists in the Arthur trace formula.

The domain D_{IV}^5 is not chosen ad hoc. It arises as the configuration space in Bubble Spacetime Theory (BST), where it independently derives the fine structure constant (0.0001%), proton–electron mass ratio (0.002%), and 23 additional physical constants with zero free parameters.

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1. Introduction

The Riemann Hypothesis — that all nontrivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = 1/2$ — has resisted proof since 1859. This paper does not prove it. Instead, we establish a precise arithmetic and spectral framework on a specific locally symmetric space and formulate a sharp conjecture that implies RH via the Arthur–Selberg trace formula.

The locally symmetric space is $\Gamma \backslash D_{IV}^5$, where $D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$ is the type-IV bounded symmetric domain of complex dimension 5, and $\Gamma = \text{SO}_0(5, 2)(\mathbb{Z})$ is the arithmetic lattice associated to the quadratic form $Q = x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - x_6^2 - x_7^2$.

What we prove:

1. Q has class number 1 (Milnor’s theorem: odd, indefinite, unimodular, rank 7).
2. Q represents all integers (Lagrange’s four-square theorem + indefiniteness).
3. $\text{Spin}(5, 2)$ satisfies strong approximation (Kneser 1966).
4. The Eisenstein series intertwining operators for the minimal parabolic are explicit products of ξ -ratios over the B_2 root system (four positive roots, multiplicities $m_\ell = 1, m_s = 3$).
5. On the standard unitary axis, the ξ -arguments have real parts 1, 3, 4, 5 — not 1/2.
6. The Cartan involution of $\text{SO}_0(5, 2)$ restricted to an embedded geodesic is exactly the functional equation reflection $s \mapsto 1 - s$.
7. The Bergman Minimum Principle: any θ -symmetric, Bergman-convex functional achieves its unique minimum at the origin ($= \text{Re}(s) = 1/2$).

What we conjecture:

The Arthur–Selberg trace formula for $\Gamma \backslash D_{IV}^5$ connects the geometric side (orbital integrals over primes, with class number 1 ensuring every local conjugacy class lifts globally) to the spectral side (where $\zeta(s)$ appears). The self-adjointness of the Bergman Laplacian, combined with the arithmetic cleanliness of class number 1, forces the ζ -zeros onto $\text{Re}(s) = 1/2$.

Why this domain. D_{IV}^5 was not selected for this purpose. It emerges as the configuration space in Bubble Space-time Theory (BST) [Koons 2026], where the substrate $S^2 \times S^1$ supports a contact graph whose automorphism group is $\text{SO}_0(5, 2)$. The same domain derives the fine structure constant ($\alpha^{-1} = 137.036, 0.0001\%$), the proton–electron mass ratio ($6\pi^5 = 1836.118, 0.002\%$), and 23 additional constants with zero free parameters. The Riemann connection is a consequence of the framework, not its motivation. The mathematical argument is self-contained: accepting BST is not required.

2. The Arithmetic of $Q = I_5 \oplus (-I_2)$

2.1 Class Number

Theorem 1 (Milnor 1958). *Every odd, indefinite, unimodular lattice of rank n is isomorphic to $I_p \oplus (-I_q)$, where (p, q) is the signature with $p + q = n$.*

Our form $Q = \text{diag}(1, 1, 1, 1, 1, -1, -1)$:

- Rank: $n = 7$
- Signature: $(p, q) = (5, 2)$
- Determinant: $\det = (+1)^5(-1)^2 = +1 \rightarrow$ unimodular
- Type: diagonal entries $\pm 1 \rightarrow$ odd (the lattice contains vectors of odd norm)
- Indefinite: has both positive and negative eigenvalues

By Milnor’s theorem: $Q \cong I_5 \oplus (-I_2)$, the *unique* lattice in its genus.

Class number = 1

This is a theorem, not a computation. One line from Milnor [1958].

2.2 Strong Approximation

Theorem 2 (Kneser 1966, Platonov 1969). *Let $G = \text{Spin}(V, Q)$ where (V, Q) is non-degenerate with $\dim V \geq 5$ and Q indefinite. Then G satisfies strong approximation with respect to any finite set of places.*

Consequence: The number of spinor genera equals 1.

Theorem 3 (Eichler 1952). *For indefinite quadratic forms of dimension ≥ 3 : class number = spinor class number.*

For our form: $\dim = 7 \geq 5$, indefinite, so $\text{Spin}(5, 2)$ satisfies strong approximation (Kneser), spinor class number = 1 (strong approximation), class number = spinor class number (Eichler). This gives an independent confirmation: class number = 1.

2.3 Universal Representation

Theorem 4. *Q represents all integers.*

Proof. Note that $Q(x) = n$ if and only if $x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 = n + x_6^2 + x_7^2$.

For $n \geq 0$: set $x_6 = x_7 = 0$. Then $x_1^2 + \dots + x_5^2 = n$. By Lagrange's four-square theorem, every non-negative integer is a sum of four squares, hence a sum of five squares (add 0^2). So n is represented.

For $n < 0$: choose x_6 large enough that $n + x_6^2 > 0$, set $x_7 = 0$, and represent $n + x_6^2$ as a sum of five squares. $\square$

In particular, Q represents every prime p . This was verified computationally for all primes up to 200 (Section 8).

2.4 Arithmetic Consequences

With class number 1 and universal representation:

1. The arithmetic lattice $\Gamma = \text{SO}_0(5, 2)(\mathbb{Z})$ is the unique lattice associated to Q . Its structure is completely determined by local data at each prime p .
2. The locally symmetric space $\Gamma \backslash D_{IV}^5$ has one connected component and no genus complications.
3. In the Selberg trace formula for $\Gamma \backslash D_{IV}^5$, every local conjugacy class (at each prime p) lifts to a global conjugacy class in Γ . No primes are “missing” from the geodesic spectrum.
4. The hyperbolic conjugacy classes are parameterized by eigenvalues of semisimple elements, determined by integer values of Q . Since Q represents all integers, every possible eigenvalue occurs.

This is the cleanest possible arithmetic setting for the trace formula.

3. The Cartan Involution and the Functional Equation

3.1 The Domain

The type-IV bounded symmetric domain is

$$D_{IV}^5 = \{z \in \mathbb{C}^5 : 1 - 2z \cdot \bar{z} + |z \cdot z|^2 > 0, |z \cdot z| < 1\}$$

with Bergman kernel $K(z, \bar{w}) = c_5 / N(z, \bar{w})^6$, where $N(z, \bar{w}) = 1 - 2z \cdot \bar{w} + (z \cdot z)(\bar{w} \cdot \bar{w})$ and $c_5 = 1920/\pi^5$. The isometry group is $\text{SO}_0(5, 2)$ with isotropy $K = \text{SO}(5) \times \text{SO}(2)$.

3.2 The Cartan Decomposition

The Lie algebra $\mathfrak{so}(5, 2)$ admits $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{m}$ with $\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$ (11 generators, the isotropy algebra) and $\mathfrak{m} \cong \mathbb{C}^5$ (10 real generators, the tangent space). The Cartan involution θ acts as $+1$ on $\mathfrak{k}$ and -1 on $\mathfrak{m}$.

On the domain, θ induces the geodesic symmetry at the origin: $\exp(X) \cdot o \mapsto \exp(-X) \cdot o$ for $X \in \mathfrak{m}$. Its fixed-point set is the origin o alone (Helgason 1978, Ch. VI).

3.3 The Critical Strip Embedding

Definition. The embedding $\iota : (0, 1) \hookrightarrow D_{IV}^5$ is defined by $\iota(\sigma) = (\sigma - 1/2) e_1$, where $e_1 = (1, 0, 0, 0, 0)$.

Under this embedding, the Cartan involution acts as:

$$\theta \circ \iota(\sigma) = -(\sigma - 1/2) e_1 = \iota(1 - \sigma)$$

This is exactly the functional equation reflection $\sigma \mapsto 1 - \sigma$, with fixed point $\sigma = 1/2$ mapping to the origin $z = 0$.

The correspondence:

Analytic	Geometric
Functional equation: $s \mapsto 1 - s$	Cartan involution: $\theta(z) = -z$
Fixed locus: $\text{Re}(s) = 1/2$	Fixed point: origin of D_{IV}^5
$ \chi(1/2 + it) = 1$	θ is an isometry at its fixed point
Critical strip $(0, 1)$	Geodesic through origin

These are the same reflection, the same fixed point, in two languages.

3.4 Isometry at the Fixed Locus

The factor $\chi(s) = \pi^{s-1/2} \Gamma((1-s)/2)/\Gamma(s/2)$ satisfies $|\chi(1/2 + it)| = 1$ for all $t \in \mathbb{R}$. Verified numerically to 50 decimal places: the critical line is the exact isometric locus.

4. Intertwining Operators and the B_2 Root System

This section contains the explicit computation. The restricted root system of $\mathfrak{so}(5, 2)$ is B_2 , with rank 2.

4.1 Root System

Let $\mathfrak{a} \subset \mathfrak{m}$ be a maximal abelian subalgebra ($\dim_{\mathbb{R}} \mathfrak{a} = 2 = \text{rank}$). The restricted roots of $(\mathfrak{g}, \mathfrak{a})$ form the root system B_2 .

Positive roots (standard choice α_1 short, α_2 long):

Root	Type	Multiplicity m_α
$\alpha_1 = e_1$	short	3
$\alpha_2 = e_2$	short	3
$\alpha_1 + \alpha_2 = e_1 + e_2$	long	1
$2\alpha_1 + \alpha_2 = 2e_1 + e_2$	long	1

The multiplicities are determined by $\dim \mathfrak{g}^{m_\alpha}$: short root multiplicity $m_s = 3$, long root multiplicity $m_\ell = 1$. This is specific to $\text{SO}_0(5, 2)$.

4.2 Half-Sum ρ

$$\begin{aligned}\rho &= \frac{1}{2} \sum_{\alpha > 0} m_\alpha \cdot \alpha = \frac{1}{2} (3\alpha_1 + 3\alpha_2 + (\alpha_1 + \alpha_2) + (2\alpha_1 + \alpha_2)) \\ &= \frac{1}{2} (6\alpha_1 + 5\alpha_2) = (3\alpha_1 + \frac{5}{2}\alpha_2)\end{aligned}$$

In coordinates: $\rho = (5/2, 3/2)$.

Check: $|\Sigma^+| = 4$ roots, total multiplicity = $3 + 3 + 1 + 1 = 8$, half-sum of multiplicities = 4. The dimension formula: $\dim G/K = 2|\rho| = 10 = \dim_{\mathbb{R}} D_{IV}^5$. ✓

4.3 The c -Function (Harish-Chandra)

For the minimal parabolic $P = MAN$ of $SO_0(5, 2)$, the Harish-Chandra c -function is:

$$c(\lambda) = \prod_{\alpha \in \Sigma^+} \frac{\xi(\langle \lambda, \alpha^\vee \rangle)}{\xi(\langle \lambda, \alpha^\vee \rangle + 1)}$$

where $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the completed Riemann zeta function and $\alpha^\vee = 2\alpha/|\alpha|^2$ is the coroot.

Writing $\lambda = s_1\alpha_1^\vee + s_2\alpha_2^\vee$ in coroot coordinates, the inner products with coroots give:

$$c(s_1, s_2) = \frac{\xi(s_1 - s_2)}{\xi(s_1 - s_2 + 1)} \cdot \frac{\xi(s_1 + s_2)}{\xi(s_1 + s_2 + 1)} \cdot \frac{\xi(2s_1)}{\xi(2s_1 + 1)} \cdot \frac{\xi(2s_2)}{\xi(2s_2 + 1)}$$

This is explicit. The completed Riemann zeta function $\xi(s)$ enters the spectral theory of D_{IV}^5 through these four ratios — one for each positive root of B_2 .

4.4 The Intertwining Operator

The Eisenstein series intertwining operator $M(s_1, s_2)$ for the minimal parabolic acts on induced representations and is given by:

$$M(s_1, s_2) = \prod_{\alpha \in \Sigma^+} \frac{\xi(\langle s, \alpha^\vee \rangle)}{\xi(\langle s, \alpha^\vee \rangle + 1)}$$

The operator $M(s_1, s_2)$ is meromorphic in (s_1, s_2) with poles and zeros determined by those of $\xi(s)$.

4.5 The Unitary Axis: An Honest Computation

The continuous spectrum of $L^2(\Gamma \backslash G)$ is parameterized by the unitary axis, where the spectral parameters take the form:

$$s_1 = \rho_1 + it_1 = \frac{5}{2} + it_1, \quad s_2 = \rho_2 + it_2 = \frac{3}{2} + it_2$$

Substituting into the ξ -arguments:

Root	ξ -argument	Value on unitary axis	Real part
$\alpha_1 - \alpha_2$	$s_1 - s_2$	$1 + i(t_1 - t_2)$	1
$\alpha_1 + \alpha_2$	$s_1 + s_2$	$4 + i(t_1 + t_2)$	4
$2\alpha_1$	$2s_1$	$5 + 2it_1$	5

Root	ξ -argument	Value on unitary axis	Real part
$2\alpha_2$	$2s_2$	$3 + 2it_2$	3

Critical observation: On the standard unitary axis, the ξ -factors are evaluated at real parts 1, 3, 4, 5 — none at 1/2. The zeros of $\zeta(s)$ at $\text{Re}(s) = 1/2$ do not appear as nodes of the spectral measure on this axis.

This means: the naive argument “self-adjointness of Δ_B constrains the continuous spectrum, which constrains ζ -zeros through the intertwining operator” does not work through the continuous spectrum of the minimal parabolic.

We state this failure explicitly because it is important. The connection between ζ -zeros and spectral constraints must go through a different mechanism.

5. The Bergman Minimum Principle

Despite the failure of the naive spectral argument, the geometric structure remains powerful.

5.1 Statement

Theorem 5 (Bergman Minimum Principle). *Let $\gamma : (-a, a) \rightarrow D$ be a geodesic through the origin of a Hermitian symmetric space D of non-compact type, with Cartan involution θ satisfying $\theta(\gamma(u)) = \gamma(-u)$. Let $F : (-a, a) \rightarrow \mathbb{R}$ satisfy:*

1. *θ -symmetry: $F(u) = F(-u)$;*
2. *Bergman convexity: F is convex in the Bergman arclength;*
3. *Properness: $F(u) \rightarrow +\infty$ as $u \rightarrow \pm a$.*

Then F achieves its unique global minimum at $u = 0$.

5.2 Proof

θ is an isometry fixing the origin. By (1), $F(u) = F(-u)$, so $F'(0) = 0$. By (2), F is convex, so $F''(0) \geq 0$. By (3), the minimum is not at the boundary. A symmetric convex proper function on a symmetric interval achieves its unique minimum at the center. $\square$

5.3 The Bergman Metric on the Geodesic

On $z = (\sigma - 1/2)e_1$, the Bergman metric is:

$$g(\sigma) = \frac{36}{(1 - (\sigma - 1/2)^2)^4}$$

This achieves its minimum $g(1/2) = 36$ at $\sigma = 1/2$ and diverges toward the Shilov boundary. The symmetry $g(\sigma) = g(1 - \sigma)$ is exact.

6. The Trace Formula Connection

The correct mechanism connecting ζ -zeros to the spectral theory of $\Gamma \backslash D_{IV}^5$ is the Arthur–Selberg trace formula — not the intertwining operators directly.

6.1 The Arthur–Selberg Trace Formula

For $\Gamma = \mathrm{SO}_0(5, 2)(\mathbb{Z})$ and f a test function on $G = \mathrm{SO}_0(5, 2)(\mathbb{R})$:

$$\underbrace{\sum_{\{\gamma\}} \mathrm{vol}(\Gamma_\gamma \backslash G_\gamma) \cdot O_\gamma(f)}_{\text{geometric side}} = \underbrace{\sum_{\pi} m(\pi) \cdot \mathrm{tr} \pi(f)}_{\text{spectral side}}$$

Geometric side: The sum runs over conjugacy classes $\{\gamma\}$ of Γ . With class number 1: - Every local conjugacy class at each prime p lifts to a global conjugacy class in Γ . - The orbital integrals $O_\gamma(f)$ for hyperbolic γ involve $\log p$ for rational primes. - Since Q represents all integers, every eigenvalue occurs — no primes are missing.

Spectral side: The sum runs over automorphic representations π of G occurring in $L^2(\Gamma \backslash G)$. This includes: - The discrete spectrum (square-integrable automorphic forms) - The continuous spectrum (Eisenstein series, where $\xi(s)$ appears) - The residual spectrum (residues of Eisenstein series)

6.2 Where $\zeta(s)$ Appears

The Riemann zeta function enters the trace formula through three channels:

1. Geometric side: The orbital integrals over hyperbolic conjugacy classes produce sums of the form $\sum_p \log p \cdot f(\log p / \log R)$, which are Dirichlet series related to ζ'/ζ .
2. Spectral side (continuous): The Eisenstein series contribution involves the intertwining operator $M(s_1, s_2)$, which is a product of ξ -ratios (Section 4.4). These contribute to the spectral side at all real parts, not just $1/2$.
3. Spectral side (residual): The poles of the Eisenstein series (where the intertwining operator has poles) give rise to the residual spectrum. The residues involve ζ -values and ζ -zeros.

The trace formula equates channels 1 and (2+3). The arithmetic cleanliness of channel 1 (class number 1, universal representation) constrains the analytic structure of channels 2 and 3.

6.3 The Key Mechanism

The fundamental point is:

The geometric side of the trace formula for $\Gamma \backslash D_{IV}^5$ sees all primes (universal representation) through a unique lattice (class number 1). The spectral side must match. The self-adjointness of Δ_B constrains the spectral side to real eigenvalues. The trace formula then constrains ζ -zeros.

The precise mechanism by which this constraint forces $\mathrm{Re}(s) = 1/2$ is the content of our conjecture.

6.4 Four Possible Mechanisms

We identify four channels through which the constraint could operate:

Mechanism A: Residual spectrum. The poles of $M(s_1, s_2)$ at ξ -zeros create residual representations. Self-adjointness constrains these to unitarity, which constrains the ξ -zeros. This requires analyzing which poles of the intertwining operator give genuine residual spectrum.

Mechanism B: Non-minimal parabolic. $\mathrm{SO}_0(5, 2)$ has maximal parabolic subgroups besides the minimal one. A different parabolic may have shifted ρ -values that place ξ -arguments at $\mathrm{Re} = 1/2$ on the unitary axis.

Mechanism C: Theta lift. The theta correspondence from $\mathrm{GL}(1)$ or $\mathrm{GL}(2)$ to $\mathrm{SO}_0(5, 2)$ lifts $\zeta(s)$ directly as an L -function attached to an automorphic representation. The lift preserves L -function zeros and embeds them in the spectral decomposition of $L^2(\Gamma \backslash G)$.

Mechanism D: Trace formula directly. Apply the trace formula with a carefully chosen test function f that isolates the contribution of ζ -zeros on the spectral side and matches it against the prime orbital integrals on the geometric side. This is Sarnak's approach in the $GL(2)$ setting and may generalize.

We believe Mechanism D is the most promising, with Mechanism C as a supporting approach.

7. The Precise Conjecture

Conjecture 1 (Trace Formula RH for $SO_0(5, 2)$). *Let $\Gamma = SO_0(5, 2)(\mathbb{Z})$ and $D = D_{IV}^5$. The Arthur–Selberg trace formula for $\Gamma \backslash D$, combined with:*

1. *Class number = 1 (every local conjugacy class lifts globally),*
2. *Universal representation (all integer eigenvalues occur),*
3. *Self-adjointness of the Bergman Laplacian Δ_B on $L^2(\Gamma \backslash D)$,*

implies that all nontrivial zeros of $\zeta(s)$ satisfy $\text{Re}(s) = 1/2$.

7.1 What the Conjecture Requires

Three steps, each using existing tools:

- (i) Orbital integrals. Compute the orbital integrals $O_\gamma(f)$ for hyperbolic $\gamma \in \Gamma$ and show they produce a sum involving $\log p$ for all primes p (enabled by universal representation). Relevant tools: Gross (1996), Gan–Hanke–Yu (2002).
- (ii) Spectral decomposition. Carry out the full spectral decomposition of $L^2(\Gamma \backslash G)$: discrete, continuous (Eisenstein), and residual. Identify which spectral contributions involve $\zeta(s)$ and at which arguments. Relevant tools: Arthur (2013), Moeglin–Waldspurger (1995).
- (iii) Matching. Show that the equality of geometric and spectral sides, together with self-adjointness, constrains the locations of ζ -zeros. This is the core step.

7.2 Potential Obstacles

We identify four ways the conjecture could fail:

1. The orbital integrals might not isolate rational primes cleanly. The bijection between hyperbolic conjugacy classes and rational primes depends on the specific arithmetic of $SO_0(5, 2)(\mathbb{Z})$. This is checkable.
2. The spectral decomposition might involve L -functions beyond ζ . The continuous spectrum for $SO_0(5, 2)$ involves L -functions for $GL(2)$ and $GL(4)$ representations, not just $\zeta = L(\cdot, 1)$. The conjecture requires that ζ can be isolated.
3. The trace formula matching might not be strong enough. The geometric side sees primes through orbital integrals; the spectral side sees zeros through L -functions. The equality constrains the relationship but might not determine zero locations uniquely.
4. The continuous spectrum complicates self-adjointness. Unlike the compact case (Selberg 1956), the non-compact quotient $\Gamma \backslash D$ has continuous spectrum. The self-adjointness of Δ_B is maintained (Langlands 1976), but extracting zero-location constraints requires careful analysis of the continuous spectral measure.

We do not claim these obstacles are trivial. We claim the question is well-posed and the tools exist.

Mechanism E: Chern critical line (added March 14, 2026). The Chern polynomial $P(h) = c(Q^5) = (1+h)^7/(1+2h)$ factors as $(h+1)(h^2+h+1)(3h^2+3h+1)$, with all non-trivial zeros on $\text{Re}(h) = -1/2$ (proved). Under

the identification $s = -h + 1/2$, this maps to $\text{Re}(s) = 1/2$. The Chern polynomial appears on the geometric side of the trace formula through the Seeley–de Witt coefficients of the heat kernel. Its critical line constrains the geometric side, which through the trace formula equality constrains the spectral side where $\zeta(s)$ lives. This is a finite-dimensional theorem that is the natural analog of RH. See `BST_Riemann_ChernPath.md` and `BST_ChernFactorization_CriticalLine.md` for the full development.

7.3 Why The Conjecture Is Plausible

Five independent lines of evidence:

- (a) The Cartan involution is exactly the functional equation reflection. This is structural, not coincidental.
- (b) The arithmetic is as clean as possible: class number 1, universal representation, strong approximation. This is the simplest setting in which to apply the trace formula.
- (c) The Selberg RH analog is proved for rank-1 symmetric spaces $(\Gamma \backslash \mathbb{H}^2)$. Totally geodesic copies of $\mathbb{H}^2$ exist inside D_{IV}^5 .
- (d) Numerical verification extends to 10^{13} zeros, all on $\text{Re}(s) = 1/2$.
- (e) The same domain independently derives 25+ physical constants with zero free parameters (Section 9).
- (f) The Chern polynomial of the compact dual Q^5 has all non-trivial zeros on $\text{Re}(h) = -1/2$ — a proved finite-dimensional analog of RH (Section 7.2, Mechanism E). The factorization $P(h) = \Phi_2(h) \cdot \Phi_3(h) \cdot (3h^2 + 3h + 1)$ is a theorem. The same functional equation symmetry ($h \mapsto -1 - h$) that forces the Chern zeros to $\text{Re}(h) = -1/2$ is the Cartan involution $s \mapsto 1 - s$ that defines the Riemann critical line. $P(1) = 42 = r \times N_c \times g$.

8. Numerical Verification

8.1 Prime Representation

All 46 primes up to 199 are explicitly represented by Q . A sample:

p	Solution $(x_1, \dots, x_5; x_6, x_7)$	$Q = p$
2	(0, 0, 0, 1, 1; 0, 0)	✓
7	(0, 1, 1, 1, 2; 0, 0)	✓
97	(0, 0, 0, 4, 9; 0, 0)	✓
199	verified	✓

All negative integers -1 to -20 are also represented. The theoretical proof (Section 2.3) guarantees all integers.

8.2 Bergman Metric

σ	$g(\sigma)$	Ratio to minimum
0.3 = 0.7	42.39	1.18×
0.4 = 0.6	37.48	1.04×
0.5	36.00	1.00 [minimum]

8.3 Isometry Verification

$|\chi(1/2 + it)| = 1.000000000000000$ for all tested t (50 decimal places). The critical line is the exact isometric locus.

8.4 Intertwining Operator Numerical Check

The ξ -function satisfies $\xi(s) = \xi(1-s)$ to machine precision. At the first Riemann zero $s = 1/2 + 14.1347i$: $\xi(s) = 0$ to 50 digits. The c -function $|c(1/2 + it)|^2$ shows dips near Riemann zero heights but (as proved in Section 4.5) these dips do not occur at $\text{Re} = 1/2$ on the unitary axis.

9. The Physical Origin of D_{IV}^5 (Brief)

For context only — the mathematical argument is independent.

D_{IV}^5 emerges from the BST framework through a forced chain:

1. S^1 is the simplest closed structure (no boundary conditions)
2. S^2 is the unique simply-connected orientable surface
3. $S^2 \times S^1$ is the substrate with intrinsic communication channel
4. $n_C = 5$ causal winding modes ($N_c = 3$ color + $N_w = 2$ weak)
5. The CR automorphism group of the contact structure is $\text{SO}_0(5, 2)$
6. Harish-Chandra: the symmetric space embeds as D_{IV}^5

From this single domain, with zero free parameters:

Quantity	BST	Observed	Precision
α^{-1}	137.036 (Wylar)	137.036	0.0001%
m_p/m_e	$6\pi^5$	1836.153	0.002%
$\sin^2 \theta_W$	3/13	0.23122	0.2% — Structural/Identified: unforced, $\text{MS-bar}(M_Z)$, runs (K1263)
Λ	$[\ln 138/50]\alpha^{56}e^{-2}$	2.89×10^{-122}	0.025%
η (baryon asymmetry)	$2\alpha^4/(3\pi)$	6.10×10^{-10}	1.4%
...	20+ <i>additional</i>		

A domain with this track record in physics deserves mathematical attention. The Riemann connection is a consequence.

10. Discussion

10.1 What This Paper Does

This paper establishes the arithmetic framework (class number 1, universal representation, strong approximation), computes the spectral machinery explicitly (B₂ root system, intertwining operators), identifies an honest obstacle (unitary axis alignment), and formulates a precise conjecture (trace formula RH for $\text{SO}_0(5, 2)$).

10.2 What This Paper Does Not Do

This paper does not prove the Riemann Hypothesis. It reduces RH to a specific question about the Arthur–Selberg trace formula on a specific locally symmetric space with the cleanest possible arithmetic. That question requires expertise in the Arthur trace formula — a different skill set from what produced the framework.

10.3 Why $SO_0(5, 2)$ and Not Another Group

Three reasons:

1. Physics. D_{IV}^5 is derived from first principles as the configuration space of the BST contact graph. It is not one of many options — it is the unique result of a forced chain of geometric choices.
2. Arithmetic. Signature $(5, 2)$ gives rank 7, odd, indefinite, unimodular — forcing class number 1 by Milnor. Lower-rank forms may not be unimodular or may have class number > 1 .
3. Root system. The restricted root system B_2 has exactly four positive roots. The ξ -function appears four times in the c -function. This is neither too simple (A_1 gives only one ξ -ratio, which is $GL(2)$ and already studied exhaustively) nor too complex to compute explicitly.

10.4 Invitation

The question is specific:

Does the Arthur–Selberg trace formula for $SO_0(5, 2)(\mathbb{Z}) \backslash D_{IV}^5$, with class number 1 on the geometric side and self-adjointness on the spectral side, force all nontrivial zeros of $\zeta(s)$ onto $\text{Re}(s) = 1/2$?

The arithmetic is established. The computation is explicit. The obstacles are identified. The question is well-posed. We invite its resolution.

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